

or ARM based processor architecture using RTOS (Real Time Operating System) or an embedded operating system is used to run a NAS server.

- **ASIC based NAS:** In this, NAS is deployed with a single ASIC chip, using hardware to implement TCP/IP and the file system.

The difference between NAS and SAN

SAN is a dedicated high-performance network for consolidated block level storage. The network interconnects storage devices, switches and hosts. High-end enterprise SANs may also include SAN directors for high performance and efficient utilisation of capacity. Servers primarily identify SAN as local attached storage, so multiple servers can share a storage pool. SANs are not dependent on the LAN and relieve pressure on the local network by offloading data directly from attached servers.

The key differences between NAS and SAN are listed below.

- **Operational environment:** NAS uses TCP/IP networks, commonly known as Ethernet, whereas SANs run on high-speed fibre channel networks, although more SANs are adopting IP-based fabrics because of the high cost of fibre networks. High performance remains a SAN requirement and flash-based fabric protocols are helping to close the gap between fibre speeds and slower IP.
- **Data processing:** NAS processes file-based data and SAN processes block data. NAS operates with a global name space and SANs have a dedicated SAN file system.
- **Protocols:** NAS connects directly to an Ethernet network via a LAN cable to network switch and utilises several protocols to connect with servers like NFS, SMB and HTTP. SAN communicates with SAN disk drive devices using the SCSI protocol, and the network is framed via SAS/SATA fabrics or iSCSI that maps over TCP/IP.
- **Performance:** NAS devices are not scalable, but high end NAS devices can scale up to petabytes using clusters. On the other hand, scalability is a major parameter for purchasing SANs. Its network architecture is highly suitable for systems administrators to scale performance and capacity.
- **Price:** NAS costs more than SAN, but is generally difficult to purchase and maintain. SANs are more complex to manage and their annual maintenance costs are more or less the same as NAS.
- **Ease of administration:** In a one-to-one comparison, NAS wins the ease of management contest. The device easily plugs into the LAN and offers a simplified management interface. SANs require more administration time than the NAS device. Deployment often requires making physical changes to the data centre and ongoing management typically requires specialised admins. The exception to the 'SAN-is-harder' argument is multiple NAS devices that do not share a common management console.
- **Use cases:** NAS is mainly used for file storage and sharing, archives management, Big Data, virtualisation and virtual desktop interface (VDI). On the other hand, SANs are used

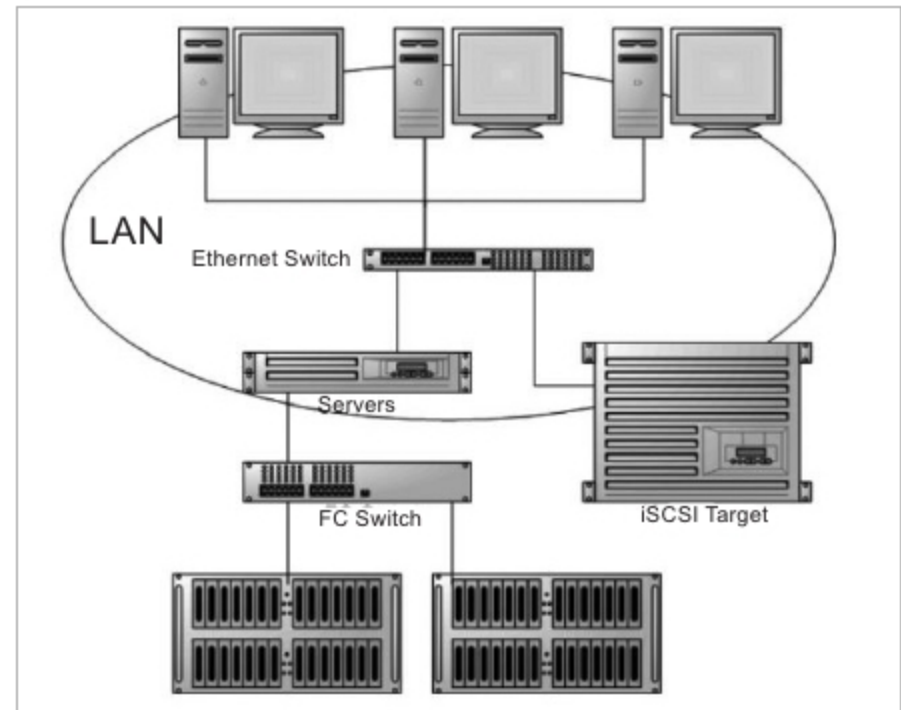


Figure 1: Typical SAN infrastructure

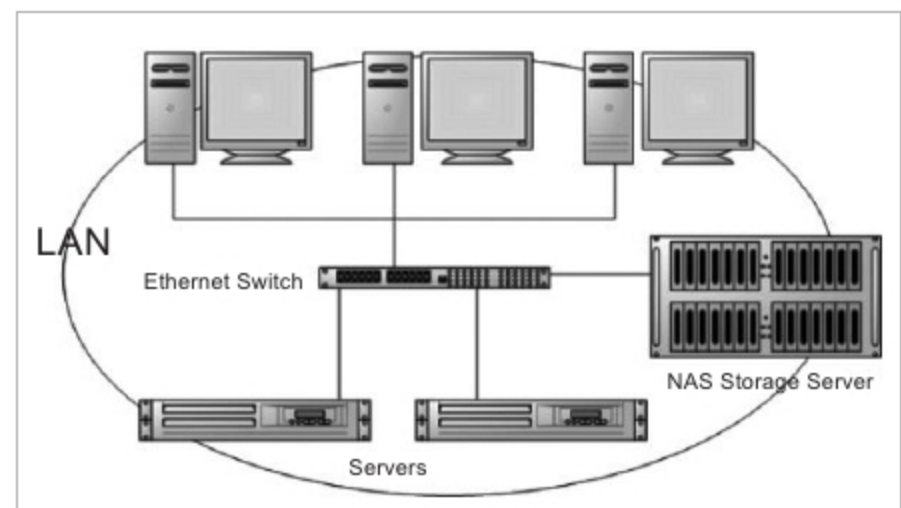


Figure 2: Typical NAS infrastructure

mostly in database management and ecommerce websites, for fast backup, virtualisation and video editing.

Advantages of NAS

- NAS entry level systems are quite inexpensive and can be purchased with storage capacities of 1TB or 2TB, which enables SMB based companies to deploy them easily. However, they can scale up very well, as there are tons of NAS appliances that support multiple arrays of disks with a storage capacity of more than 1000TB.
- NAS systems support storage of data as files, and support both CIFS (common Internet file system) and NFS protocols, which can be accessed via TCP/IP based Ethernet networks and facilitate access for multiple users.
- NAS technology supports many RAID levels, ranging from RAID 0 and RAID 1 to RAID 5, to make sure that individual disk failures don't result in data loss.
- NAS appliances come with easy GUI interfaces that can be accessed and administrated from remote locations over TCP networks including the Internet/VPNs/leased lines.
- Most NAS devices support their own cloud service, which allows users to remotely access or even synchronise data on devices with an Internet connection. The sharing can